

Simulation of hydrated DMPC phospholipid membrane

Author: Maitane Fariñas Argoitia

Facultat de Física, Universitat de Barcelona, Diagonal 645, 08028 Barcelona, Spain.

I. INTRODUCTION

The study of biological membranes and their interactions with water molecules is of great importance in understanding the fundamental processes underlying various biological phenomena. Phospholipid bilayers are integral components of cellular membranes, serving as barriers that regulate the exchange of ions and molecules between cells and their environment. The behaviour of water molecules within these lipid bilayers plays a crucial role in maintaining the structural integrity and functionality of cellular membranes.

In this practical session, we explore the dynamics of hydrated DMPC phospholipid membranes through molecular simulations. By obtaining trajectories of both the lipids and water molecules, we aim to analyse the rotational dipolar autocorrelation function of the water molecules along the simulation-time for different temperatures and distinct regions of the bilayer (determined by the lipid-water distances). The molecular dynamics simulations have been performed using the simulation package NAMD, designed to run simulations of biomolecules in aqueous systems.

The rotational dipolar autocorrelation function provides insights into the rotational motion and hydrogen bonding dynamics of water molecules in the vicinity of the lipid bilayer. By calculating this function, we gain insights into how water molecules near the lipid bilayer behave compared to those located further away. Repeating this process for a set of temperatures enables us to explore the influence of temperature on water dynamics within the membrane system.

In summary, this practical session focuses on investigating the rotational dipolar autocorrelation function of water molecules in hydrated DMPC phospholipid membranes. By examining different sets of water molecules in relation to temperature and lipid proximity, we uncover important insights into the behaviour of water within the membrane system. This knowledge advances our understanding of membrane-water interactions in biological systems and has implications for studying membrane stability and functionality.

II. SYSTEM DEFINITION

The system we are aiming to simulate is a hydrated DMPC phospholipid membrane. The membrane consists of several key components that contribute to its structure and functionality:

1. **Phospholipids:** these components are the primary building blocks of the membrane. They are a class of lipids whose molecule has a hydrophilic

head, containing a phosphate group, and two hydrophobic *tails*, derived from fatty acids, joined by an alcohol residue.

2. **Lipid Bilayer:** The DMPC molecules form a bilayer arrangement; they self-assemble due to the hydrophobic effect: the hydrophilic heads of the phospholipids lie on the exterior part of the membrane, facing the surrounding water; the hydrophobic tails are oriented toward the interior, creating the hydrophobic core of the lipid bilayer. This arrangement is essential for the stability and functionality of the membrane. It acts as a barrier, controlling the entry and exit of molecules and ions into and out of the membrane.
3. **Hydration Shell:** Surrounding the phospholipid bilayer, is a layer of water molecules known as the hydration shell. Water molecules interact with the hydrophilic head groups of the phospholipids, creating a water-rich region at the membrane's surface. This hydration shell plays a vital role in maintaining the membrane's integrity and facilitating interactions with solutes.

Our total systems consists of 128 DMPC phospholipids, arranged in a bilayer form, and 4377 water molecules, located at both sides of the lipid bilayer. A snapshot of the membrane at the end of the equilibration run can be seen in Fig. 1.

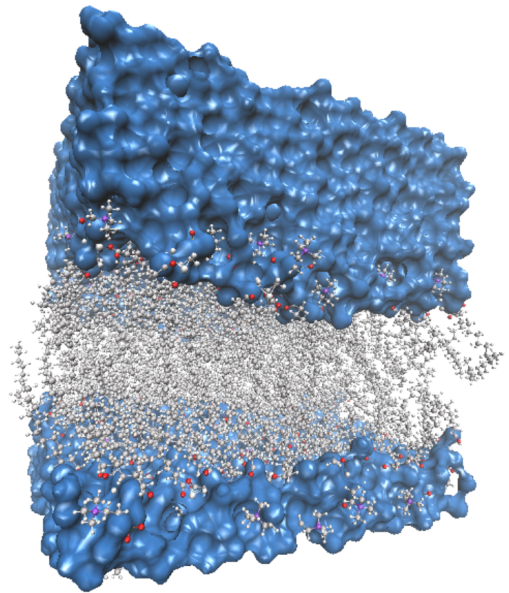


Fig. 1: Snapshot of the hydrated DMPC phospholipid membrane at the end of the equilibration run, for $T = 296$ K.

III. SIMULATIONS

The NAMD simulation package has been utilised to conduct the necessary molecular dynamic simulations in order to achieve the desired data for the analysis. NAMD is a highly parallelised code which allows us to solve the dynamics of big systems efficiently.

For each thermodynamic configuration ($p = 1$ atm and the chosen T from our temperature set), we will run two simulations.

1. **Equilibration run:** To achieve the desired thermodynamic state, an equilibration simulation is conducted, wherein the system is gradually brought to equilibrium from its initial conditions: A NPT simulation with fixed temperature and pressure, where the system mimics the experimental conditions.
2. **Production run:** Then, a production simulation run will be performed in order to obtain the trajectories of the lipids and the water molecules: these simulations will initiate from the equilibrium configuration, and will be a NVT-ensemble simulation, with fixed simulation-box and temperature, allowing the membrane size not to change over time.

In order to explore both the temperature dependence and the lipid-water distance dependence on the autocorrelation function of the water dependence, we chose a set of four temperatures: $T = 290$ K, $T = 296$ K, $T = 310$ K and $T = 300$ K. The three chosen ranges of varying distance from the lipid bilayer are $d \leq 3\text{\AA}$, $3\text{\AA} < d < 9\text{\AA}$ and $d \geq 9\text{\AA}$.

IV. ANALYSIS

As previously stated, the aim of this project is to study the rotational dipolar autocorrelation function of the water molecules of the system in different scenarios.

The rotational dipolar autocorrelation function, $C_{rot}(t)$ is used to analyse the rotational dynamics of molecules. It describes the time-dependent correlation between the orientation of a molecular dipole at different points in time. It quantifies how the orientation of a molecule's –water molecules, in this study– dipole moment changes over time, and it provides information about the rotational motion and relaxation of the molecule.

The rotational dipolar autocorrelation function is defined by

$$C_{rot}(t) \equiv \frac{\langle \hat{\mu}(t) \cdot \hat{\mu}(0) \rangle}{\langle \hat{\mu}(0) \cdot \hat{\mu}(0) \rangle} \quad (1)$$

where $\hat{\mu}$ is the water dipole vector at time t and $\langle \dots \rangle$ denotes the average over all water molecules. Fig. 2 shows how $\hat{\mu}$ is defined.

The resulting $C_{rot}(t)$ curves will be shown and discussed in the following section.

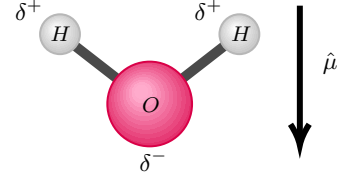


Fig. 2: Representation of the dipole-moment of a water molecule.

In order to filter the water molecules that belong to each of the lipid-hydration distance ranges, we have used the VMD visualisation program. Water molecules have been filtered to the different distance-ranges using conditions of distances (`within d of lipid`), as well as conditions for selecting molecules only if all three atoms lie within said distance range (`within 1 of numbonds=2`). This way, some of the molecules might have been left out from all three ranges, so the sum of the total number of water molecules of each range will not equal the number of water molecules of the system. However, this way we can really make sure that no single hydrogen nor oxygen atom is taken into account in the calculation of $\hat{\mu}$.

V. RESULTS AND DISCUSSION

In this section we will show the results obtained by the analysis of the trajectories for different temperatures and different hydration shells.

A. Temperature dependence

In this first section, we will focus on analysing the dependence of the rotational dipolar autocorrelation function on temperature. We will study whether the temperature has an impact on the shape of $C_{rot}(t)$, and what its effect is.

Fig. 3 shows the autocorrelation functions of the water molecules of the whole membrane for different temperatures.

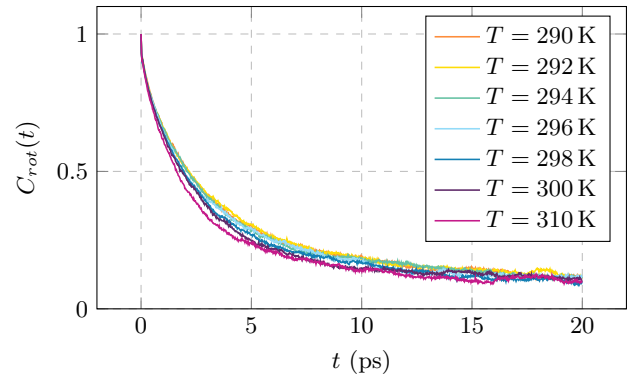


Fig. 3: Rotational dipolar autocorrelation function in time, for different temperature values.

Fig. 3 shows the expected rotational dipolar autocorrelation function for the water molecules, as well as the

expected tendency for said function for varying temperatures.

At the start of the simulation, the initial orientation of the dipole and the orientation at time t are highly correlated. Such correlation decays as the simulation evolves, which means that the orientation of the dipoles become less influenced and dependent on its past values, as time lag increases.

We can see that the decay is faster for larger temperature values; thermal fluctuations play an important role in that, since higher temperatures enhance molecular motion and increase the overall kinetic energy of the system. The increased thermal energy disrupts the persistence of dipole orientations, leading to a quicker loss of correlation over time. However, for $t > 15 \text{ ps}$, it seems that the dependency on temperature is somewhat lost, as the $C_{rot}(t)$ curves get mixed together, around a similar value.

B. Lipid-hydration shell distance dependence

In this section we will analyse how the distance between the lipid and the different hydration shells affects the orientation of the dipole over time.

In order to do so, we plotted the rotational dipolar autocorrelation function for different lipid-shell distances, for different temperature values. The following four pictures show

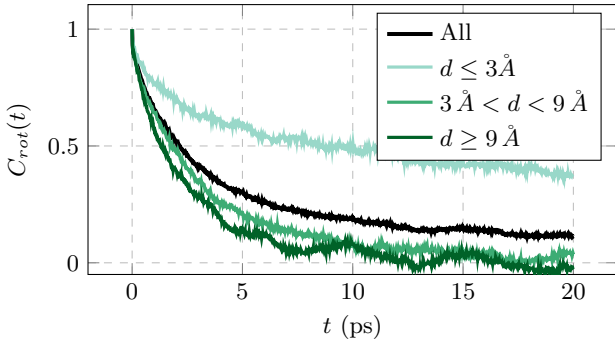


Fig. 4: Rotational dipolar autocorrelation function of the water molecules at $T = 290 \text{ K}$.

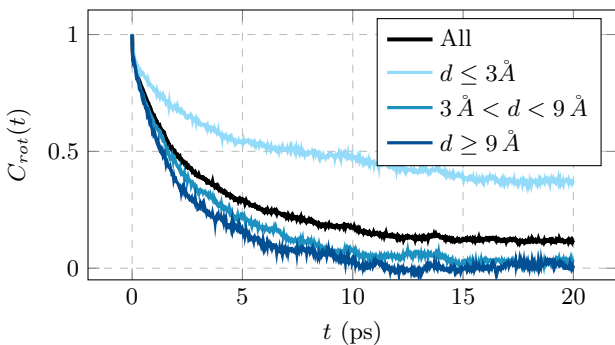


Fig. 5: Rotational dipolar autocorrelation function of the water molecules at $T = 296 \text{ K}$.

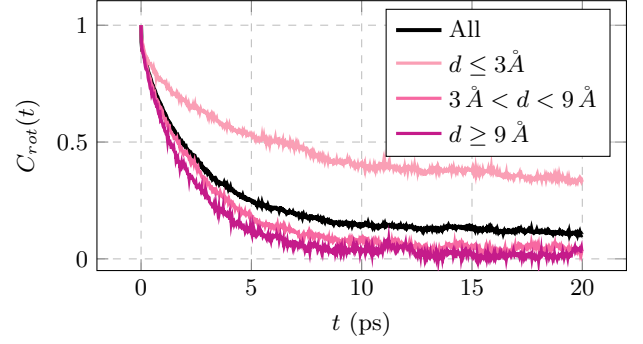


Fig. 6: Rotational dipolar autocorrelation function of the water molecules at $T = 300 \text{ K}$.

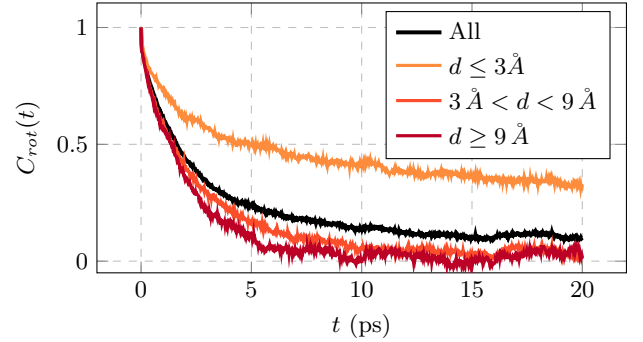


Fig. 7: Rotational dipolar autocorrelation function of the water molecules at $T = 310 \text{ K}$.

Fig. 4, 4, 4 and 4 show the rotational dipolar autocorrelation function as a function of the distance from the lipid to the water molecules. All four temperatures show the same tendency: the two outermost shells have fast decaying autocorrelation functions (specially the $d \geq 9 \text{ Å}$ shell), whereas the closest shell to the lipid decays much slower, having almost a value of $C_{rot} = 0.5$ at 20 ps time. This behaviour can be attributed to the influence of intermolecular interactions. As molecules move further away from the lipid, the strength of these interactions decreases. Consequently, the rotational motion becomes less influenced by neighbouring molecules, leading to faster decay and lower values of the autocorrelation function over longer times.

The intermolecular interaction between water molecules and the lipid membrane are actually between the water and the head of the phospholipid, as it is the head the one with hydrophilic nature. These interaction primarily involve hydrogen bonding.

In the case of a DMPC phospholipid, the hydrophilic head group typically contains a phosphate group and a choline group. The oxygen atoms in the phosphate group and the nitrogen atoms in the choline group can act as hydrogen bond acceptors, while the hydrogen atoms in water molecules act as hydrogen bond donors. These hydrogen bonds help stabilise the interaction between the water and the phospholipid.

The effective range of the hydrogen bonding interaction is usually of a few angstroms, typically less than 4 Å . This can be clearly seen in the previous plots: a big

difference in the autocorrelation function can be seen between the closest hydration shell ($d \leq 3 \text{ \AA}$) and the two furthest ($3 \text{ \AA} < d < 9 \text{ \AA}$ and $d \geq 9 \text{ \AA}$), which means that the hydrogen bonds only have an effect on the dipole orientation of the water molecules at $d \leq 3 \text{ \AA}$, while for biggest distances, the effect of thermal fluctuations win.

C. Lipid-hydration shell distance and temperature dependence

Finally, in order to study how the temperature affects the strength of the intermolecular interactions depending on the lipid-water distance, we will analyse the three different shells one by one. In Figures 8, 9 and 10 the autocorrelation functions corresponding to the three ranges are shown.

The inner shell is the one that shows the biggest dependence on temperature out of the three: the difference between the curves is the biggest out of the three cases, and, although we might have expected lower temperatures to have the highest autocorrelation through the whole simulation, and vice-versa, the relationship is almost maintained, meaning that, even if thermal fluctuations are able to dim the effect of intermolecular bonds (hydrogen bonds) at longer times, it takes longer to do so for lower temperatures. The middle shell starts the simulation by having the lowest temperatures have the highest autocorrelation, but by around 15 ps , all curves are mixed together at almost $C_{rot} = 0$. That is the case for the outermost shell as well: at around 10 ps , we are no longer able to distinguish the effect of temperature in the curves, and all curves have reached the zero value, which means that the thermal fluctuations are able to overcome the intermolecular bonds completely. We need to take into account that the results in the simulations also vary from simulation to simulation because the equations employed are Langevin equations, which are of stochastic nature.

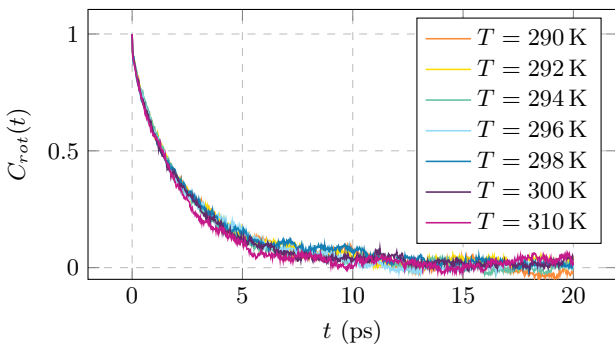


Fig. 8: Rotational dipolar autocorrelation function of the outermost shell.

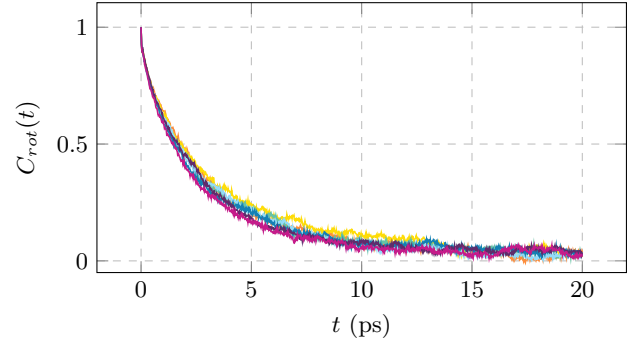


Fig. 9: Rotational dipolar autocorrelation function of the middle shell.

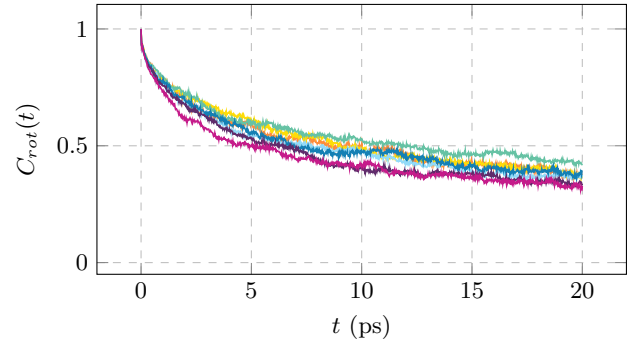


Fig. 10: Rotational dipolar autocorrelation function of the innermost shell.

VI. CONCLUSIONS

In this computational project we have simulated a hydrated DMPC phospholipid membrane, which has allowed us to study the effects of the temperature and the lipid-water layer distance on the orientation of the dipole moment of the water molecules over time, by computing the rotational dipolar autocorrelation function.

We have seen that temperature does play a role in the dipolar autocorrelation function of the water molecules, specially in the inner shell, where the autocorrelation function decays faster for higher temperatures, due to the thermal fluctuations being able to dim the hydrogen bonding interactions between the head of the phospholipid (hydrophilic) and the water molecules.

However, it is the lipid-water layer distances what have bigger effect on the autocorrelation function. Due to the range of the hydrogen bonding interactions, only the water molecules of the innermost shell are affected by the lipid membrane; the dipole orientation of the molecules on this layer have a higher correlation, since the interaction makes them take certain values of the dipole moment. However, for distances bigger than $3 \text{ \AA}$, the strength of the bonding decays very fast, and therefore, the molecules in the middle and outermost shell lose dipolar correlation faster.